

Measuring cell volume

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V2).

Measuring cell volume

1. Transfer 25 μL of culture to a 1.5 mL microcentrifuge tube.
2. Macroscopic strains: sample using 100 μL wide-bore tips, then break clusters into small pieces using 100 μL regular-bore tips to facilitate single-layer crushing later (see [[Breaking up and crushing clusters: collected methods]]).
3. Pellet clusters ($5,000 \times g$, 1 min), wash with 1 mL $1 \times \text{PBS}$.
4. Stain in 100 μL of 5 μM Calcofluor White in $1 \times \text{PBS}$ at room temperature.
5. Crush 5 μL of stained clusters into a single layer of cells between a slide and coverslip, without shearing.
6. Image on a Nikon Eclipse Ti inverted microscope, brightfield and UV/fluorescent channels, ≥ 5 fields of view per strain at $\times 40$ magnification. Set focus in brightfield to the plane where cells appear grey, then move down 0.7 μm for sharp fluorescent cell-wall signal.
7. Cell volume is calculated as an ellipsoid: $V = 4/3 \times \pi ab^2$, where a and b are the major- and minor-axis lengths measured from segmented cells.

Image analysis pipeline (cell segmentation): same pipeline as [[Measuring cell aspect ratio]].